

Technical Document #628: Cloud Computing - Comprehensive Overview

Technical Document #628: Cloud Computing - Comprehensive Overview

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Content delivery networks (CDNs) distribute content globally to reduce latency.
- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.